Day 14 Homework: Does a fact-checker earn its place in the team?

Name: Sadasivam Mithran Date: 2026-10-15 Repository: https://github.com/mithransadasivam/FDE (folder **HW #14**)

This document has the three things the homework asks for: the fact-checker (its prompt, its handoff model and its tests passing), the comparison output on five topics, and the decision.

1. The fact-checker

Where: `app/fact_checker.py` (new file). The fact-checker does not change `app/team.py`: it is added through build_team()'s four extension points (extra_nodes, route_fn, state_schema, feedback_fn) in `build_team_with_fact_checker()`. (Before starting I also completed the four unfinished Day 14 class TODOs in handoffs.py, team.py and notice_checks.py, which the team needs in order to run at all.)

Handoff model

```
class FactCheck(BaseModel):
    """Fact-checker -> supervisor (and, through the supervisor, the writer)."""

    claims_checked: int = Field(ge=0, description="How many checkable claims the notice makes")
    unsupported: list[str] = Field(
        default_factory=list,
        description="Each claim the notes do not support: the quoted claim, then what is wrong",
    )
```

Prompt

```
FACT_CHECKER_PROMPT = """You are the fact-checker in an IT communications team. You get research notes
(the only facts we have) and a draft IT notice. Check every claim in the notice against the notes.

1. List every checkable claim in the title, body and actions: times, dates, numbers, causes, what is
   affected, who is affected, what staff must do, and promises about what happens next.
   Set claims_checked to how many you found.
2. A claim is supported when a fact in the notes says it. Rewording is fine.
3. A claim is unsupported when no fact in the notes says it, or when it goes further than the notes
   (a made-up time, a guessed cause, a number of people, an estimated fix time, a promise).
4. For each unsupported claim write one line: the claim in quotes, then what is wrong with it.
   Quote the notice exactly. If every claim is supported, leave unsupported empty.
5. Do not judge style, tone or length: that is the reviewer's job. Only check facts."""
```

Routing: the class rule first

```
def route_with_fact_check(state: dict) -> str:
    """The class rule first. Only where it would send a draft to the reviewer do we step in."""
    nxt = route(state)
    if nxt != "reviewer":
        return nxt
    fc = checked_this_draft(state)
    if fc is None:
        return "fact_checker"
    if fc["unsupported"] and (state.get("revisions") or 0) < MAX_REVISIONS:
        return "writer"
    return "reviewer"
```

Every new draft goes to the fact-checker before the reviewer. If it lists unsupported claims the draft goes back to the writer (with those claims as feedback) while revisions < MAX_REVISIONS. The check is stored with the draft it was made on, so a new draft is always checked again.

Tests passing (five, using the Day 14 fakes)

```
tests/test_my_fact_checker.py::test_fact_check_returns_a_checked_handoff PASSED [ 20%]
tests/test_my_fact_checker.py::test_a_bad_reply_is_repaired_once_then_fails PASSED [ 40%]
tests/test_my_fact_checker.py::test_an_unsupported_claim_goes_back_to_the_writer_before_review PASSED [ 60%]
tests/test_my_fact_checker.py::test_a_clean_draft_goes_straight_to_the_reviewer PASSED [ 80%]
tests/test_my_fact_checker.py::test_the_route_calls_the_class_rule_first_and_stops_after_max_revisions PASSED [100%]
============================== 5 passed in 2.12s ===============================
```

The tests cover: a checked FactCheck handoff; a bad reply repaired once, then failing; an unsupported claim sent back to the writer **before** the reviewer (the reviewer never sees it); a clean draft going straight to the reviewer; and the route calling the class rule first and stopping after MAX_REVISIONS. The whole suite also passes: 314 tests.

2. The comparison (compare_fact_checker on five topics)

Command: `python -m scripts.compare_fact_checker --topics data/my_notice_topics.csv`. Topics: the 3 class topics plus 2 of my own (PRN-310 stuck print jobs; WIFI-102 Wi-Fi failures after a password change). Real model calls; nothing is sent anywhere.

Honest note on the runs. The first full run is shown below. The researcher (shared by both versions) sometimes returns JSON that does not parse, and the team then stops with no notice. This hit the checker version on VPN and Wi-Fi in run 1, so I re-ran those topics (both versions). On VPN, the team without the checker finished only 1 of 3 attempts and the team with it 2 of 3. This is the researcher, not the checker. An earlier attempt crashed on an OpenRouter rate limit (429) and counted 0 tokens; I fixed both in the Day 6 model wrapper `app/rag.py`.

Output of run 1 (first full run)

```
The Finance Portal outage this morning  [team]  5 calls, 12460 tokens, 83.3 s, revisions 0
  problems: none found
  title: Finance Portal Outage

The Finance Portal outage this morning  [team + checker]  11 calls, 18126 tokens, 160.3 s, revisions 2
  problems: missing: status page
  title: Finance Portal outage

VPN error VPN-809 for people returning from leave  [team]  7 calls, 15518 tokens, 124.2 s, revisions 1
  problems: none found
  title: VPN error VPN-809 for returning staff

VPN error VPN-809 for people returning from leave  [team + checker]  5 calls, 14147 tokens, 148.8 s, revisions 0
  problems: no notice

Delays to external email in Company Mail today  [team]  7 calls, 15201 tokens, 71.6 s, revisions 1
  problems: none found
  title: Delays in external email delivery

Delays to external email in Company Mail today  [team + checker]  8 calls, 16071 tokens, 72.0 s, revisions 1
  problems: none found
  title: Delays in external email delivery in Company Mail

Print jobs stuck in the queue with error PRN-310  [team]  9 calls, 17297 tokens, 78.3 s, revisions 2
  problems: missing: Print Queue
  title: Print jobs stuck with error PRN-310

Print jobs stuck in the queue with error PRN-310  [team + checker]  11 calls, 22608 tokens, 132.9 s, revisions 2
  problems: none found
  title: Print jobs stuck with PRN-310 error

Wi-Fi sign-in failures with error WIFI-102 after a password change  [team]  10 calls, 18561 tokens, 152.3 s, revisions
     2
  problems: none found
  title: Wi-Fi sign-in failures with error WIFI-102 after password change

Wi-Fi sign-in failures with error WIFI-102 after a password change  [team + checker]  5 calls, 12473 tokens, 72.0 s, r
    evisions 0
  problems: no notice

version         calls   tokens  seconds  problems
team               38    79037    509.7         1
team + checker     40    83425    586.0         3

The notices are saved in logs/team_comparison.jsonl. Read them: the checks cannot judge tone or truth.
```

Output of the re-runs (Wi-Fi + VPN, then VPN again)

```
VPN error VPN-809 for people returning from leave  [team]  5 calls, 13911 tokens, 150.0 s, revisions 0
  problems: no notice

VPN error VPN-809 for people returning from leave  [team + checker]  11 calls, 18953 tokens, 83.0 s, revisions 2
  problems: missing: self-service portal
  title: VPN error VPN-809 for devices off over 30 days

Wi-Fi sign-in failures with error WIFI-102 after a password change  [team]  9 calls, 17089 tokens, 60.6 s, revisions 2
  problems: none found
  title: Wi-Fi Sign-In Issue After Password Change

Wi-Fi sign-in failures with error WIFI-102 after a password change  [team + checker]  13 calls, 22014 tokens, 82.3 s,
    revisions 2
  problems: none found
  title: Wi-Fi sign-in failures with error WIFI-102

version         calls   tokens  seconds  problems
team               14    31000    210.6         1
```

```
team + checker     24    40967    165.3         1

The notices are saved in logs/team_comparison.jsonl. Read them: the checks cannot judge tone or truth.
VPN error VPN-809 for people returning from leave  [team]  5 calls, 14124 tokens, 45.6 s, revisions 0
  problems: no notice

VPN error VPN-809 for people returning from leave  [team + checker]  11 calls, 19327 tokens, 91.0 s, revisions 2
  problems: none found
  title: VPN Certificate Expiration Issue (Error VPN-809)

version         calls   tokens  seconds  problems
team                5    14124     45.6         1
team + checker     11    19327     91.0         0

The notices are saved in logs/team_comparison.jsonl. Read them: the checks cannot judge tone or truth.
```

3. Results and decision

Name: Mithran Sadasivam Date: 2026-10-15

The fact-checker

• What it checks (one sentence): every checkable claim in the draft notice (times, numbers, causes, who is affected, what staff must do, promises) against the research notes, and it lists each claim the notes do not support.

• Where it sits in the team, and where its findings go: between the writer and the reviewer (`app/fact_checker.py`). If it lists unsupported claims, the supervisor sends the draft back to the writer with those claims as feedback, while `revisions < MAX_REVISIONS`; a clean draft (or one where the revisions are used up) goes on to the reviewer.

• Its prompt's most important line, and why: "A claim is unsupported when no fact in the notes says it, or when it goes further than the notes (a made-up time, a guessed cause, a number of people, an estimated fix time, a promise)." Writers invent in exactly these ways, so naming them makes the checker look for them. The line "Do not judge style" keeps it from doing the reviewer's job.

The comparison

Topics file used: `data/my_notice_topics.csv` (the 3 class topics plus 2 of my own: PRN-310 stuck print jobs, and WIFI-102 Wi-Fi failures after a password change).

The model is not deterministic and the shared model account allows 20 requests a minute, so the first run needed re-running for VPN and Wi-Fi (see "What went wrong with the runs" below). Each row below is a pair that finished in the **same** run, except VPN (see the note under the table).

Topic	Version	Calls	Tokens	Seconds	Revisions	Automatic problems	Unsupported claims you found by reading
Finance Portal outage	team	5	12,460	83.3	0	0	1: "This is due to a 503 error" (the notes say what a 503 means, not that it is the cause)
	team + checker	11	18,126	160.3	2	1 (missing: status page)	0, but the actions line is off topic and the notice lost "follow the status page"
VPN-809 (returning from leave)	team	7	15,518	124.2	1	0	0
	team + checker	11	19,327	91.0	2	0	1: "there are no open tickets related to this issue" (the help desk data has 5 unresolved VPN-809 tickets)
Company Mail delays	team	7	15,201	71.6	1	0	1: "This affects all staff sending emails to external recipients"
	team + checker	8	16,071	72.0	1	0	0
Print jobs, PRN-310 (own)	team	9	17,297	78.3	2	1 (missing: Print Queue)	0
	team + checker	11	22,608	132.9	2	0	0
Wi-Fi, WIFI-102 (own)	team	9	17,089	60.6	2	0	1: "there are no open tickets related to this issue" (INC-1032 and INC-1012 are open)
	team + checker	13	22,014	82.3	2	0	2: "use the 'Fix Wi-Fi' tool in the IT portal" (invented tool; the article says the device management app); "No further updates are planned unless the situation changes"

Totals over the four topics where both versions finished in the same run (Finance, Mail, Print, Wi-Fi):

team 30 calls, 62,047 tokens, 293.8 s. Team + checker 43 calls, 78,819 tokens, 447.5 s.

Automatic problems 1 vs 1. Unsupported claims found by reading (these four topics): 3 vs 2.

Totals, all five topics (VPN taken as team from run 1, team + checker from run 3):

team 37 calls, 77,565 tokens. Team + checker 54 calls, 98,146 tokens.

Extra cost (four matched topics): +13 calls (+43%), +16,772 tokens (+27%), +153.7 s (+52%). With VPN included: +46% calls, +27% tokens.

What went wrong with the runs (reported, not hidden): 6 attempts at the VPN topic: the team without the checker finished only 1 of 3 (the other two stopped with "researcher handoff failed: not valid JSON") and the team with the checker finished 2 of 3 (one stopped the same way). Wi-Fi failed once for the checker version for the same reason. This is the researcher, which both versions share, so it is not the checker's doing and it is not an argument for or against it. A likely cause is the 300-token reply limit cutting the researcher's JSON off; I did not change that. Raw logs: `docs/evidence/team_comparison_all_runs.jsonl` (all 16 rows, every notice), and `logs/compare_fact_checker_output_*.txt`.

Also: the very first attempt crashed on an OpenRouter 429 (rate limit) and showed 0 tokens. I fixed both in `app/rag.py` (the Day 6 model wrapper: it now passes token counts on and waits and retries on a 429). `app/team.py` was not changed for the fact-checker.

What it caught, and what it missed

• One unsupported claim the checker caught (quote it): in a traced Finance Portal run (`logs/fact_check_trace.txt`) the writer's second draft said "Several users have reported this issue." The checker flagged it: "No fact in the notes mentions user reports." The writer removed it. In the same trace it also caught "IT is working on the issue." and, in the Company Mail run, "Updates will be posted on the service status page." (no fact says so), and these were fixed before the reviewer saw the draft.

• One problem it missed or flagged wrongly:

  • **Missed:** in the WIFI-102 notice that passed the checker, the writer told staff "On managed laptops, use the 'Fix Wi-Fi' tool in the IT portal". No fact mentions that tool (KB-4401 says "the device management app"), and the same notice says "No further updates are planned unless the situation changes", a promise that is also not in the notes. I found both only by reading. It was also inconsistent: in the Finance trace it flagged "affecting all users" in draft 1 but accepted "This affects all staff who use the portal" in draft 3.

  • **Flagged wrongly (too strict):** in the Print trace it flagged "Some print jobs are stuck in the queue with error PRN-310" because the notes only give ticket counts. That is a fair reading of the notes, and it costs a revision for little gain.

  • **Cannot catch by design:** "there are no open tickets related to this issue" appeared in the VPN and Wi-Fi notices. It contradicts the help desk data, but the researcher's notes said it, so the checker (which only compares the draft with the notes) agreed.

Decision

Keep the fact-checker? **Only for some notices: keep it for notices sent to all staff, skip it for low-stakes internal drafts.** It costs about 43% more calls, 27% more tokens and 52% more time over four topics, and the automatic checks scored 1 problem for both versions, so on those checks alone it earns nothing. What it does is remove made-up details a person would otherwise have to find (user reports, promises of updates, "IT is working on it") before review, and a wrong detail in a notice to everyone costs far more than 27% more tokens. But it is a first filter and not a guarantee: it missed an invented tool and a promise, flagged some claims too strictly, and cannot see errors in the research itself, so the person approving must still read every notice and check the facts the notes got wrong (for example, "no open tickets" when there are open ones).

Appendix: what the checker said (trace of three team + checker runs)

Produced by `python -m scripts.show_fact_checks "<topic>"`. It prints the research facts, each draft, and what the fact-checker flagged.

```
######## The Finance Portal outage this morning
RESEARCH FACTS:
  - [service_status] The Finance Portal is currently in outage status since 07:58.
  - [find_tickets] There are 3 open tickets related to the Finance Portal outage.
  - [KB-6601] A 503 response means the portal's application servers are unavailable, usually during an outage or maintenar
  - [KB-6601] Individual users cannot fix a 503 error; they should follow the status page.
  - [KB-6601] Raising individual tickets during a known outage slows the response.
  - [[1] Day06_Slide15_policy_it_support_slas.pdf, page 1] The IT Support and Service Level Policy explains how to get IT
    port and how quickly the service desk responds.
  - [[2] Day06_Slide15_policy_it_support_slas.pdf, page 1] P1 means a critical service is down for many users.

DRAFT 1: Finance Portal currently unavailable
  The Finance Portal is currently unavailable (since 07:58) with a 503 error, which means the servers are down. This is a
    incident affecting all users. IT is working on the issue.
  actions: ['Do not raise individual tickets about this outage - it slows the response.', 'Check the service status page f
    updates.']
  FACT-CHECK: 5 claims, unsupported: ['"This is a P1 incident affecting all users." - The notes confirm P1 means a critica
    ervice is down for many users, but do not explicitly state it affects all users.', '"IT is working on the issue." - Th
    otes do not mention IT\'s current actions or status.']

DRAFT 2: Finance Portal currently unavailable
  The Finance Portal is currently unavailable since 07:58, showing a 503 error (servers unavailable). Several users have r
    rted this issue. Do not raise individual tickets for this known outage as it slows response times.
  actions: ['Check the service status page for updates', 'Do not raise individual tickets for this issue']
  FACT-CHECK: 5 claims, unsupported: ['Several users have reported this issue: No fact in the notes mentions user reports.

DRAFT 3: Finance Portal is currently unavailable
  The Finance Portal is currently unavailable due to an outage that started at 07:58. This affects all staff who use the p
    al. A 503 error means the portal's servers are unavailable, which happens during outages or maintenance.
  actions: ['Do not raise individual tickets about this outage, as it slows the response.', 'Check the service status page
    r updates.']
  FACT-CHECK: 5 claims, unsupported: []
  REVIEW: {'verdict': 'revise', 'issues': ['The notice does not explicitly state that there is nothing staff can do to fix
    e 503 error.', 'The notice does not mention when the next update will come or where to find it.', 'The notice does not
    plicitly state that the issue is classified as P1, which is a critical service down for many users.', 'The notice does
    t explicitly state that there are 3 open tickets related to the Finance Portal outage.']}
######## Print jobs stuck in the queue with error PRN-310
RESEARCH FACTS:
  - [service_status] The Print Queue service is currently operational.
  - [find_tickets] There are 4 open tickets related to the Print Queue, 3 of which mention error PRN-310.
  - [KB-3301] PRN-310 means a job is waiting for a printer that is offline or out of paper.
  - [KB-3301] When many users on one floor see PRN-310, the print server's spooler service usually needs restarting.
  - [KB-3301] A single user's stuck job can be cancelled and resent.
  - [[1] Day10_Slide21_policy_printing_scanned.pdf, page 2] Report printer faults through a P3 ticket in the IT portal, in
    ding the printer name shown on the label.

DRAFT 1: Print Jobs Stuck with Error PRN-310
  Some print jobs are stuck in the queue with error PRN-310, which means a job is waiting for a printer that is offline or
    t of paper. The Print Queue service is currently operational. If your job is stuck, cancel it and resend it to another
    inter. Report printer faults through a P3 ticket in the IT portal, including the printer name shown on the label. We w
     provide an update if the situation changes.
  actions: ['Cancel and resend stuck print jobs to another printer.', 'Report printer faults through a P3 ticket in the IT
    rtal, including the printer name.']
  FACT-CHECK: 6 claims, unsupported: ['"We will provide an update if the situation changes." - This is a promise not menti
    d in the notes.', '"Some print jobs are stuck in the queue with error PRN-310" - The notes mention 3 tickets related t
    RN-310, but do not quantify how many print jobs are stuck.']

DRAFT 2: Print jobs stuck with error PRN-310
  Some staff are experiencing print jobs stuck in the queue with error PRN-310. PRN-310 means a job is waiting for a print
    that is offline or out of paper. The Print Queue service is currently operational.
  actions: ['Cancel and resend your print job if it is stuck.', 'Report printer faults through a P3 ticket in the IT porta
    including the printer name shown on the label.']
  FACT-CHECK: 5 claims, unsupported: ['"Some staff are experiencing print jobs stuck in the queue with error PRN-310" - Th
    otes mention 3 tickets related to error PRN-310 but do not specify that \'some staff\' are experiencing it.', '"Cancel
    d resend your print job if it is stuck." - The notes mention that a single user\'s stuck job can be cancelled and rese
     but do not specify that this applies to all staff or all stuck jobs.']

DRAFT 3: Print jobs stuck with error PRN-310
  Some print jobs are stuck in the queue with error PRN-310, which means a job is waiting for a printer that is offline or
    t of paper. The Print Queue service is currently operational.
  actions: ['Report printer faults through a P3 ticket in the IT portal, including the printer name shown on the label.']
  FACT-CHECK: 4 claims, unsupported: []
  REVIEW: {'verdict': 'revise', 'issues': ["The notice does not specify who is affected (e.g., 'some staff' or 'users on o
    floor').", 'The notice does not mention when the next update will come or where to find it.', "The notice does not inc
    e the fact that a single user's stuck job can be cancelled and resent.", "The notice does not mention the potential so
    ion of restarting the print server's spooler service when many users on one floor see PRN-310."]}
######## Delays to external email in Company Mail today
RESEARCH FACTS:
  - [service_status] Company Mail service is currently degraded, with external email delays of up to 30 minutes.
  - [find_tickets] There are 4 open tickets related to Company Mail, including issues like external email delays and mailb
```

```
    quotas.
  - [[1] Day06_Slide15_policy_data_handling.pdf, page 1] Confidential information must not be emailed outside the company
    ess encrypted, and restricted information must never be emailed.
  - [KB-2202] During service degradation, external messages can be queued for up to an hour, and no action is needed from
    rs.

DRAFT 1: Delays to external email in Company Mail today
  Company Mail is currently experiencing delays of up to 30 minutes for external emails. No action is needed from users. U
    tes will be posted on the service status page.
  actions: ['Do not email confidential information outside the company unless encrypted.', 'Never email restricted informa
    n outside the company.']
  FACT-CHECK: 5 claims, unsupported: ['"Updates will be posted on the service status page." - No fact in the notes mention
    pdates being posted on the service status page.']

DRAFT 2: Delays to External Email in Company Mail Today
  Company Mail is currently experiencing delays of up to 30 minutes for external emails. No action is needed from users du
    g this time. Updates will be provided as more information becomes available.
  actions: ['Do not email confidential information outside the company unless encrypted.', 'Never email restricted informa
    n outside the company.']
  FACT-CHECK: 5 claims, unsupported: ['Updates will be provided as more information becomes available. - This is a promise
    t supported by the notes.']

DRAFT 3: Delays to external email in Company Mail
  Company Mail is currently experiencing delays of up to 30 minutes for external emails. During service degradation, exter
     messages can be queued for up to an hour. Confidential information must not be emailed outside the company unless enc
    ted, and restricted information must never be emailed.
  actions: ['No action is needed from users.', 'Do not email confidential information outside the company unless encrypted
     'Never email restricted information.']
  FACT-CHECK: 5 claims, unsupported: []
  REVIEW: {'verdict': 'revise', 'issues': ['The first sentence should combine what is happening and who is affected, per s
    e guide rule 1.', 'The notice does not specify when the next update will come or where to find it, per style guide rul
    .', 'The body is 60 words, which is under the 180-word limit, but the structure could be more concise per style guide
    es 1 and 2.', 'The notice includes policy reminders that are not directly related to the current issue, which may conf
     the main message, per style guide rule 2.']}
```